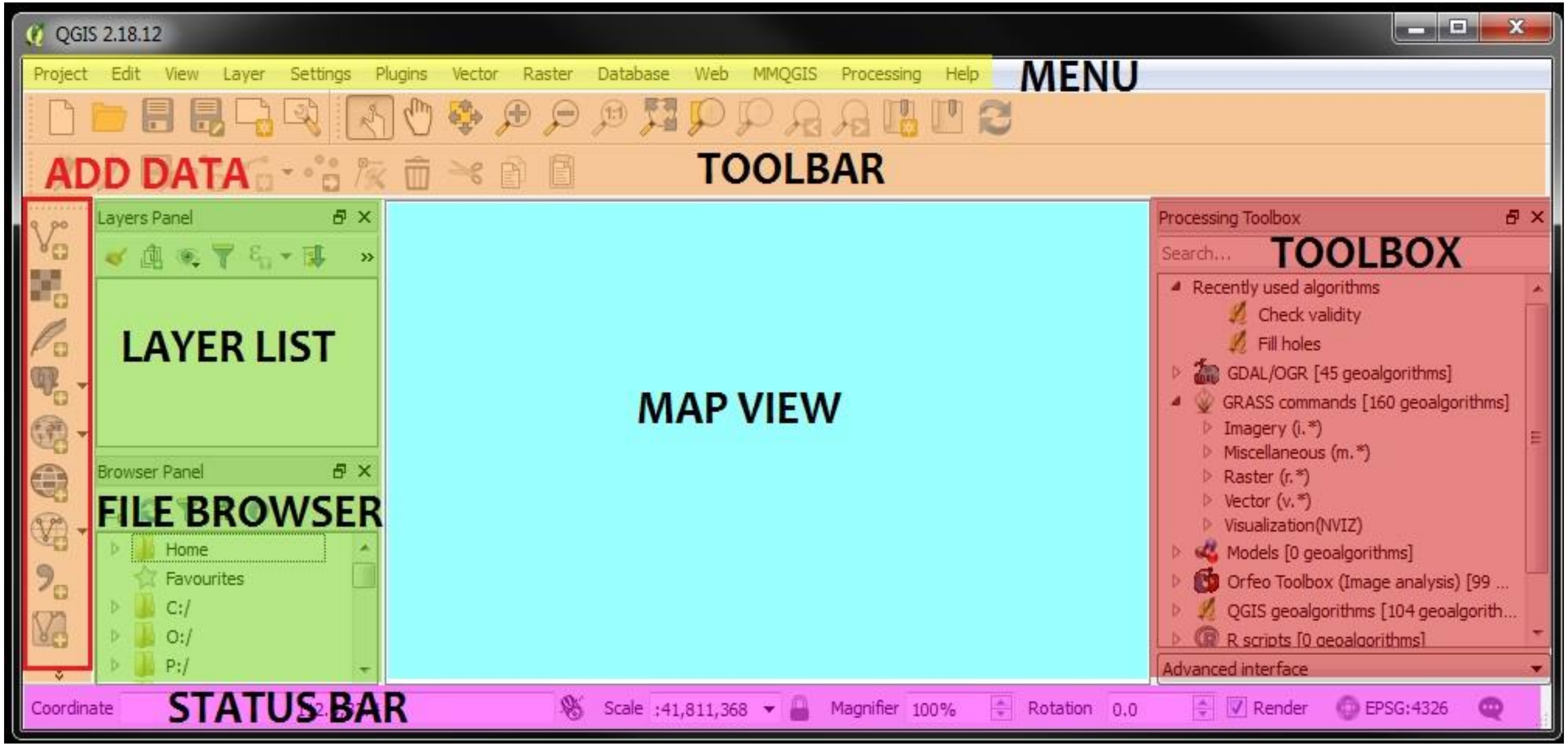
ICT 4 GEOMATICS

LABORATORY of GIS

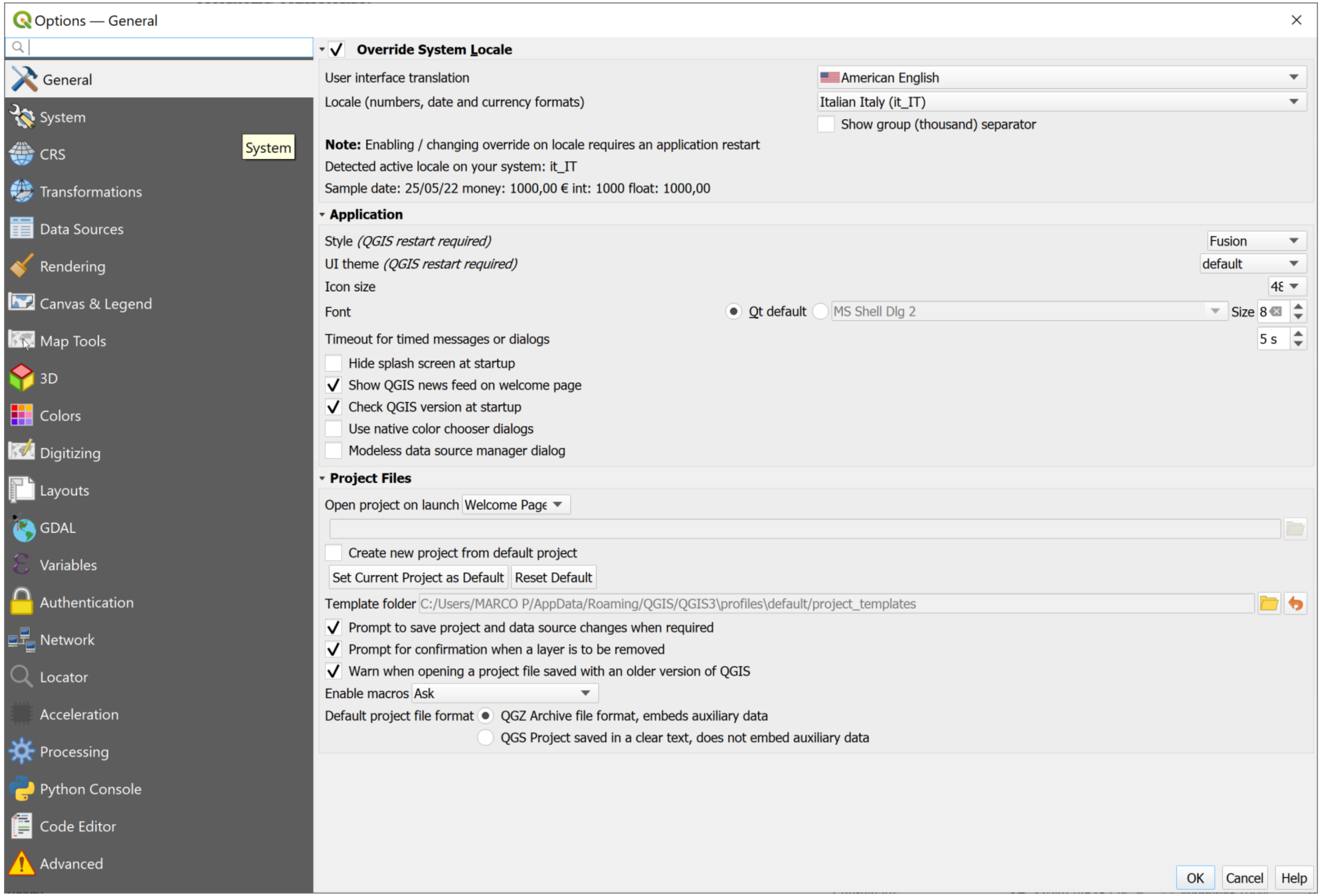
Prof. MARCO PIRAS

QGIS interface

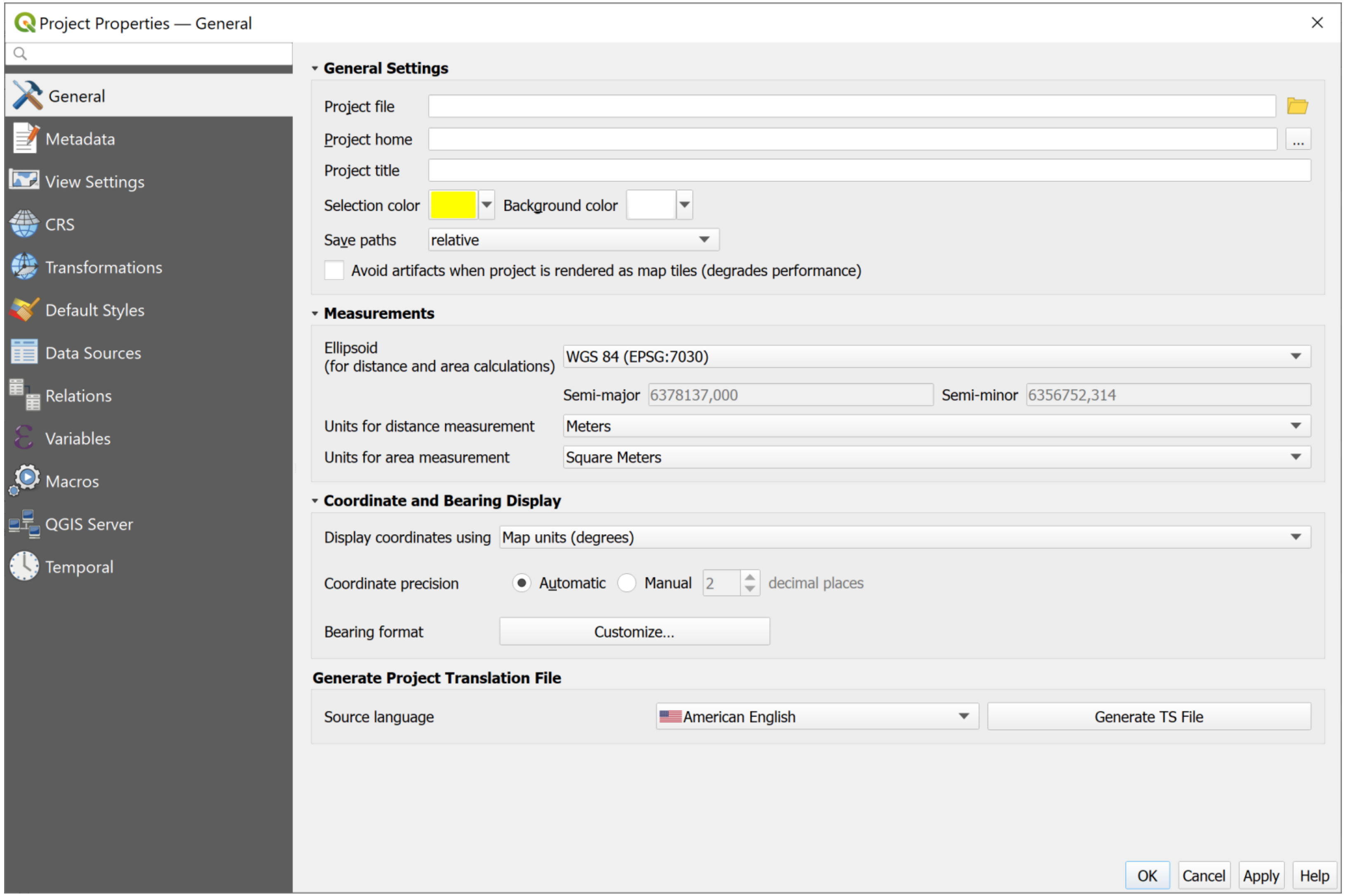


https://docs.qgis.org/2.14/en/docs/training_manual/

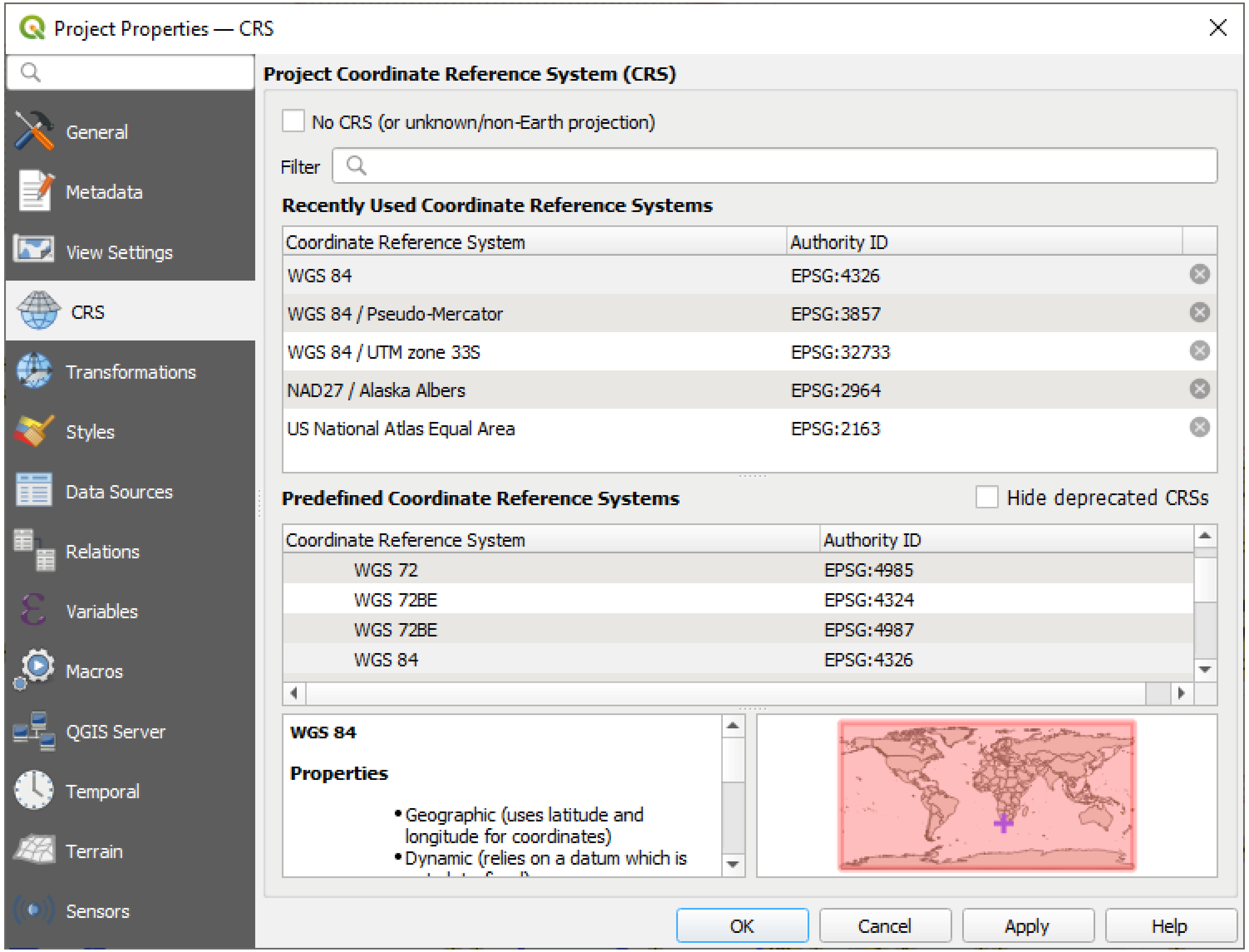
Setting → options



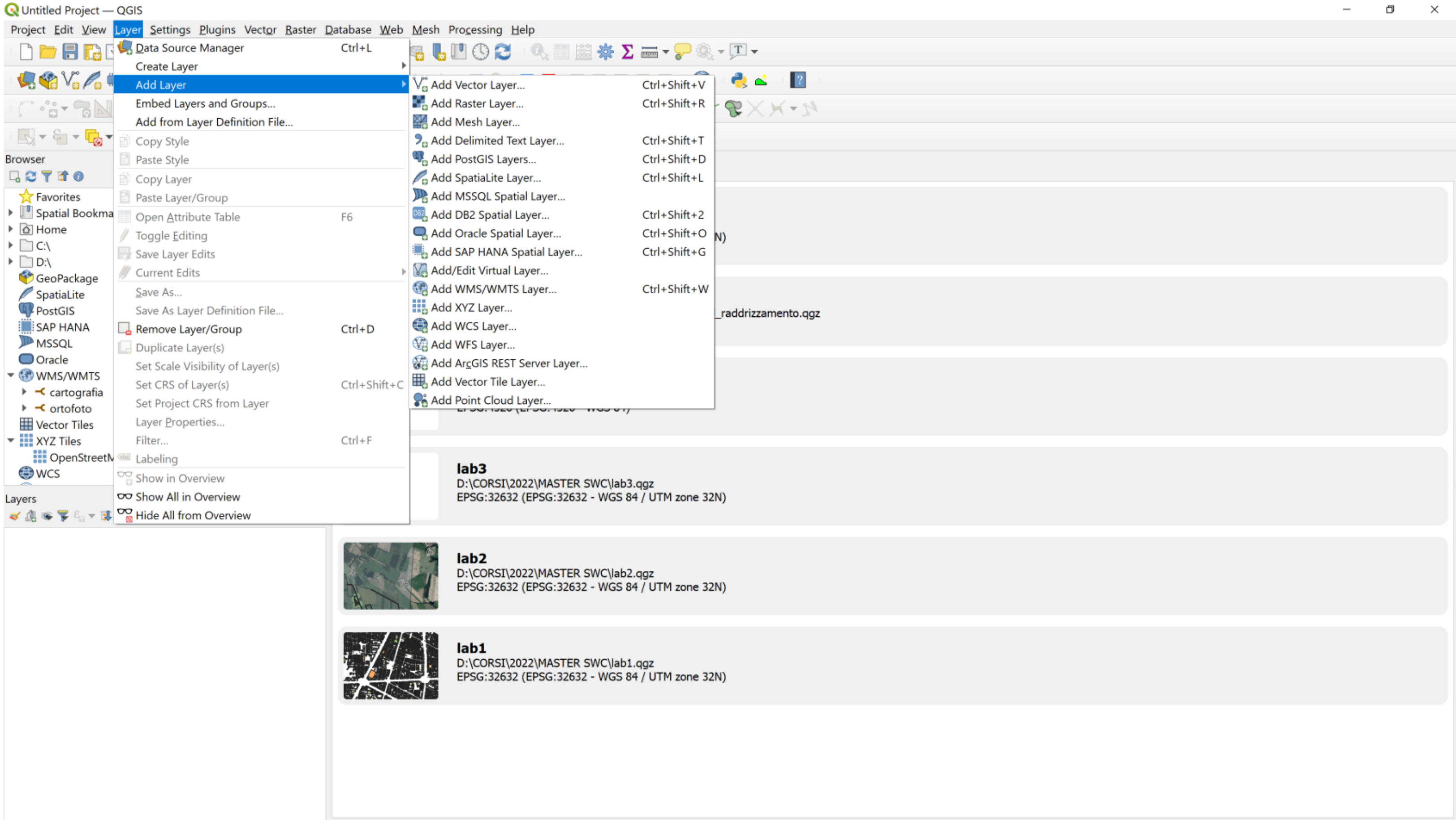
Project → properties



Reference system



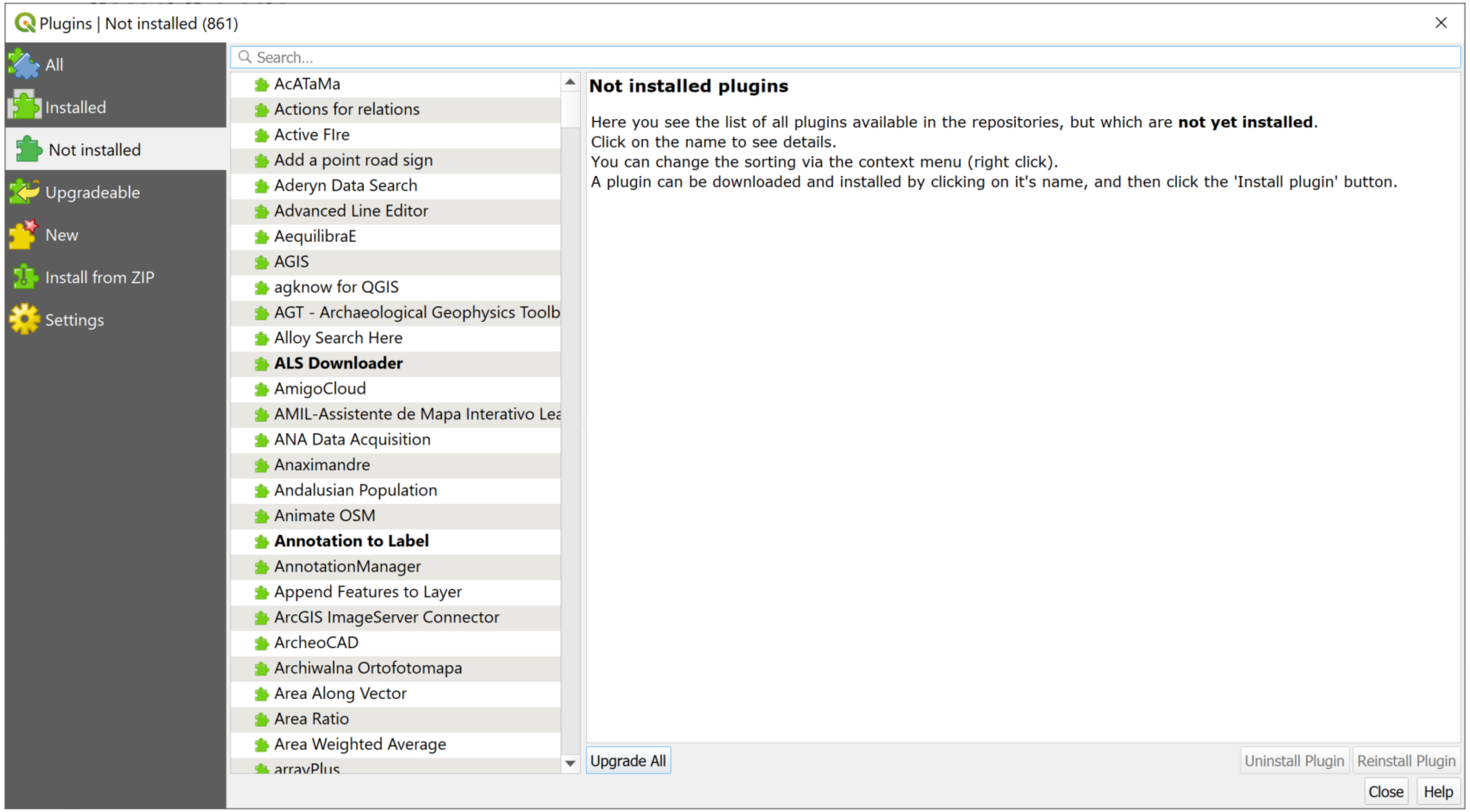
Data import


Untitled Project — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Data Source Manager Ctrl+L
Create Layer
Add Layer
Embed Layers and Groups...
Add from Layer Definition File...
Copy Style
Paste Style
Copy Layer
Paste Layer/Group
Open Attribute Table F6
Toggle Editing
Save Layer Edits
Current Edits
Save As...
Save As Layer Definition File...
Remove Layer/Group Ctrl+D
Duplicate Layer(s)
Set Scale Visibility of Layer(s)
Set CRS of Layer(s) Ctrl+Shift+C
Set Project CRS from Layer
Layer Properties...
Filter... Ctrl+F
Labeling
Show in Overview
Show All in Overview
Hide All from Overview
Add Vector Layer... Ctrl+Shift+V
Add Raster Layer... Ctrl+Shift+R
Add Mesh Layer...
Add Delimited Text Layer... Ctrl+Shift+T
Add PostGIS Layers... Ctrl+Shift+D
Add SpatiaLite Layer... Ctrl+Shift+L
Add MSSQL Spatial Layer...
Add DB2 Spatial Layer... Ctrl+Shift+2
Add Oracle Spatial Layer... Ctrl+Shift+O
Add SAP HANA Spatial Layer... Ctrl+Shift+G
Add/Edit Virtual Layer...
Add WMS/WMTS Layer... Ctrl+Shift+W
Add XYZ Layer...
Add WCS Layer...
Add WFS Layer...
Add ArcGIS REST Server Layer...
Add Vector Tile Layer...
Add Point Cloud Layer...
Browser
Favorites
Spatial Bookma
Home
C:\
D:\
GeoPackage
SpatiaLite
PostGIS
SAP HANA
MSSQL
Oracle
WMS/WMTS
cartografia
ortofoto
Vector Tiles
XYZ Tiles
OpenStreetM
WCS
Layers
_raddrizzamento.qgz
lab3
D:\CORSI\2022\MASTER SWC\lab3.qgz
EPSG:32632 (EPSG:32632 - WGS 84 / UTM zone 32N)
lab2
D:\CORSI\2022\MASTER SWC\lab2.qgz
EPSG:32632 (EPSG:32632 - WGS 84 / UTM zone 32N)
lab1
D:\CORSI\2022\MASTER SWC\lab1.qgz
EPSG:32632 (EPSG:32632 - WGS 84 / UTM zone 32N)

plugins

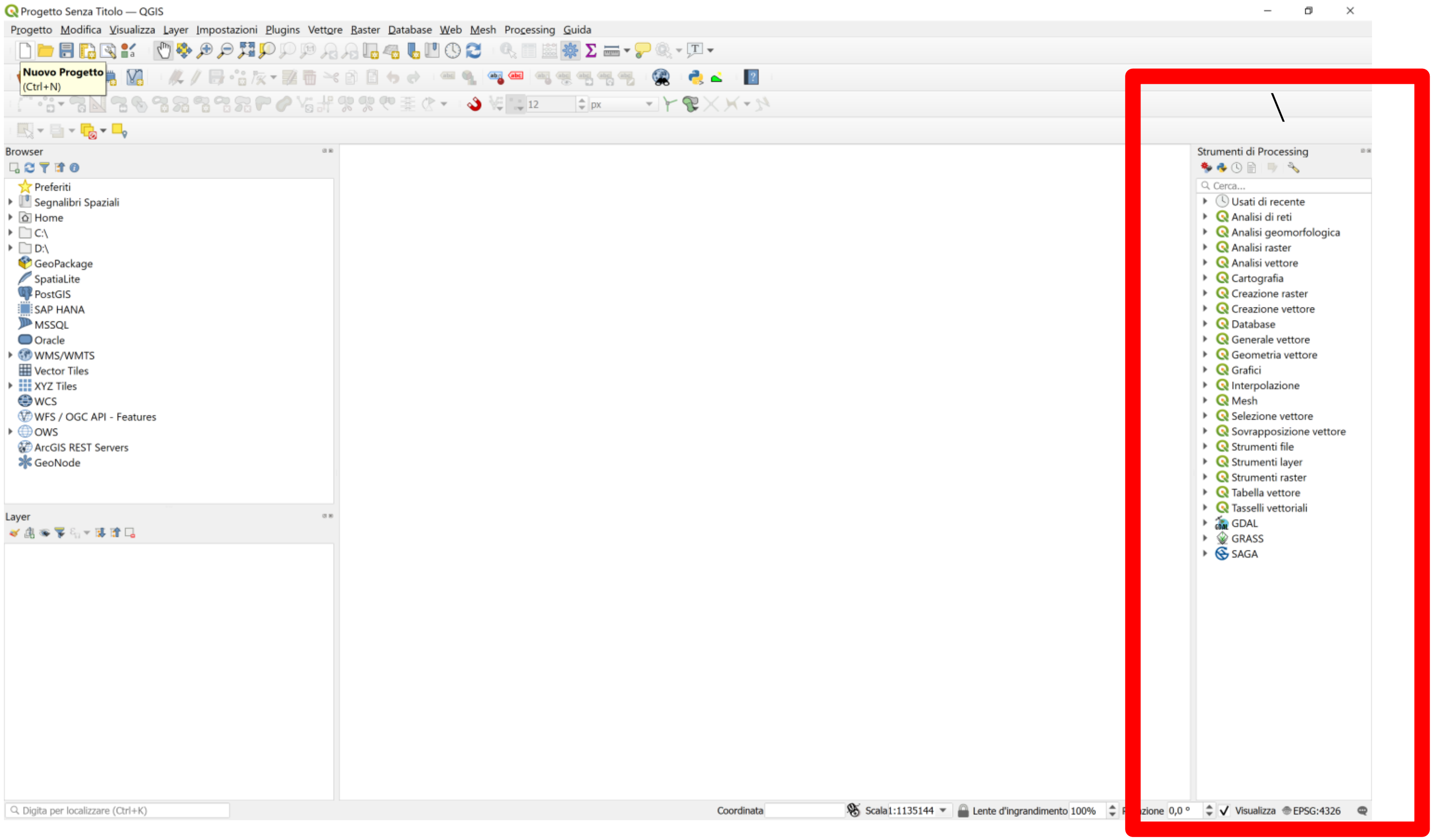
MENU → PLUGINS

BE careful about the version



Processing tools

MENU → PROCESSING



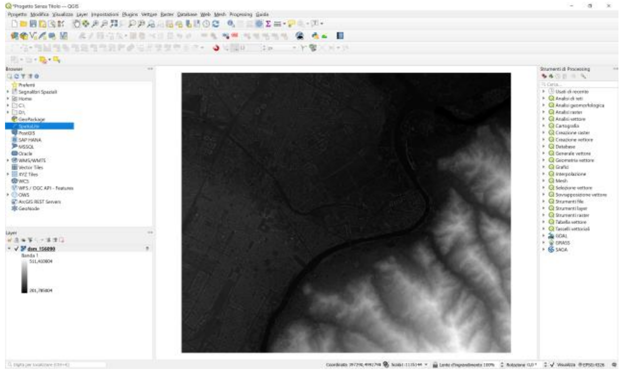
EX 1

AIM:
IMPORT A RASTER AND VECTOR DATA and CREATE A NEW LAYER

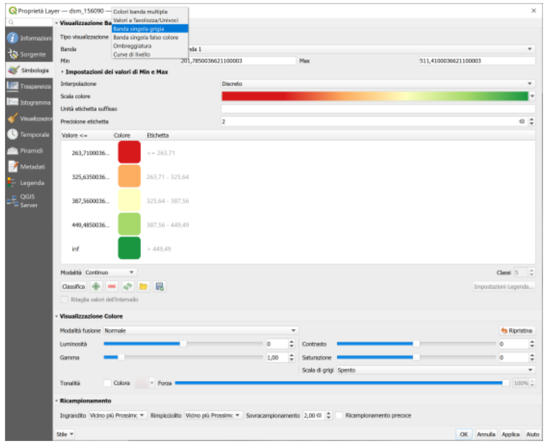
DATA:
1) RASTER dtm_155120.asc
2) VECTOR: FOLDER «IMM» OF BDTRE REGIONE PIEMONTE (scale 1:10000)

Ex 1

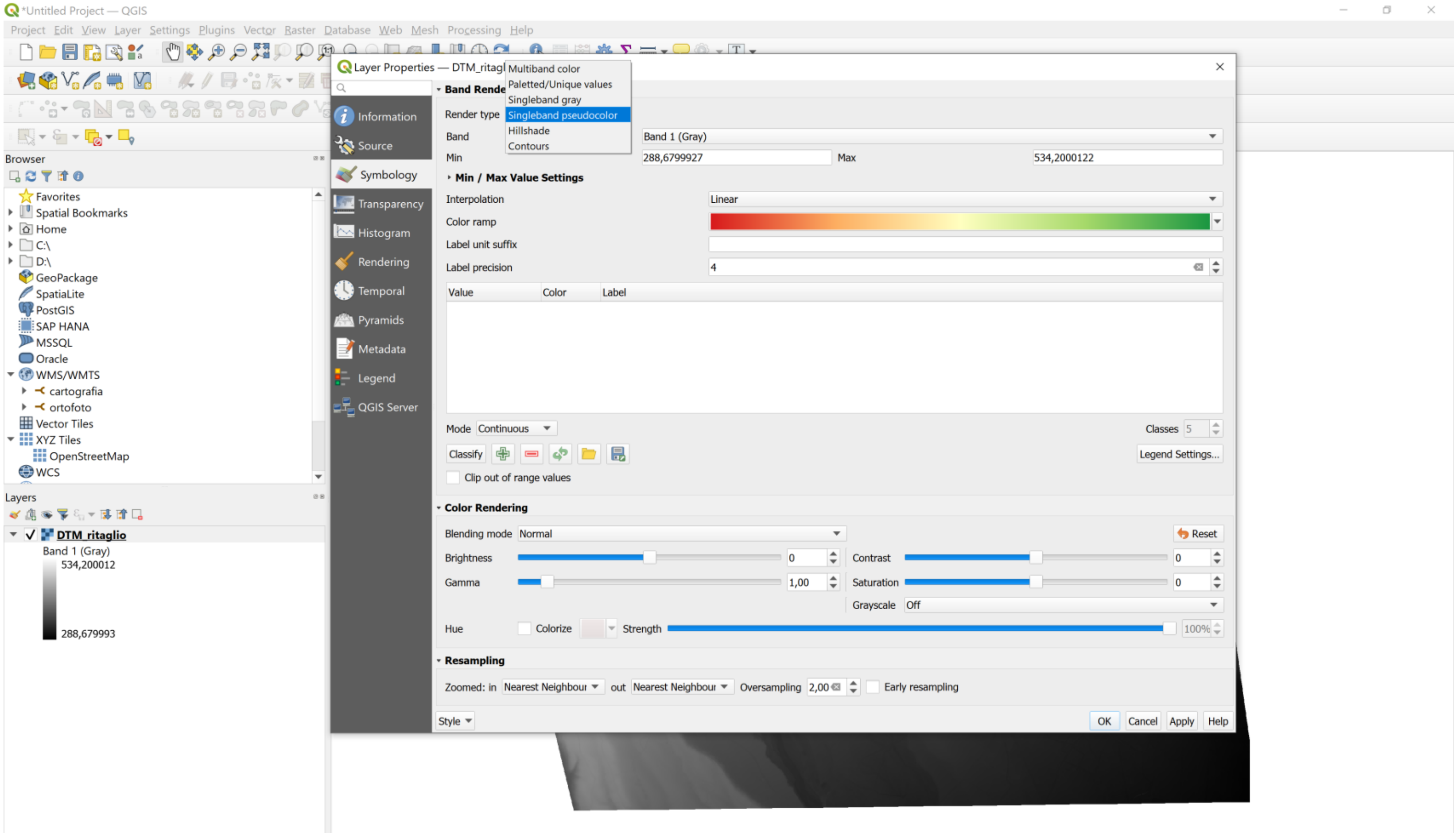
1) Select «LAYER» → ADD LAYER → ADD RASTER LAYER
2) Select the file dtm_155120.asc
3) CLICK ADD
4) SELECT LAYER DSM
5) Click RB (right button) and select properties
6) Select «symbology» and select singleband pseudocolor



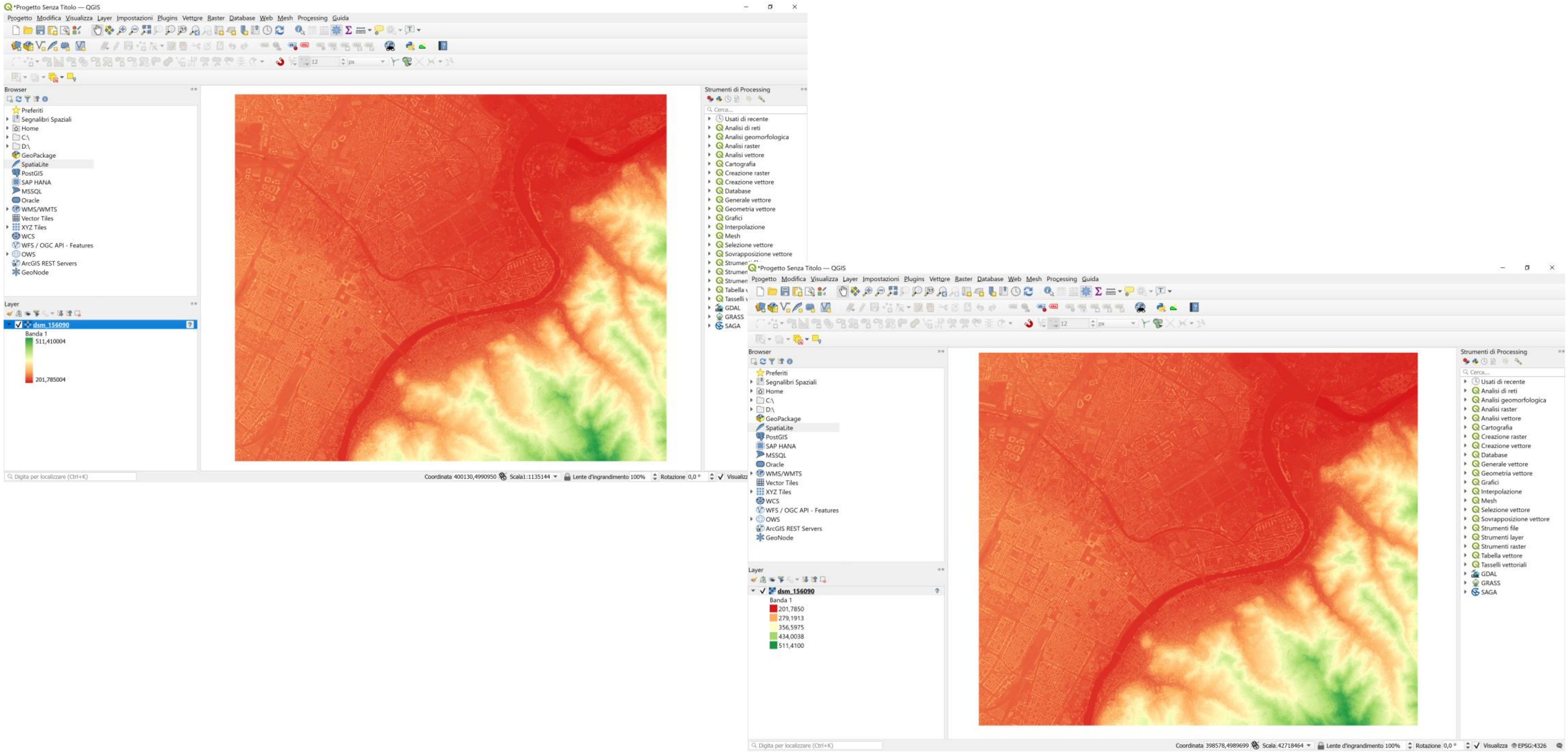
7) Select the option of the visualization
8) Classify
9) Select color ramp
10) Click ok



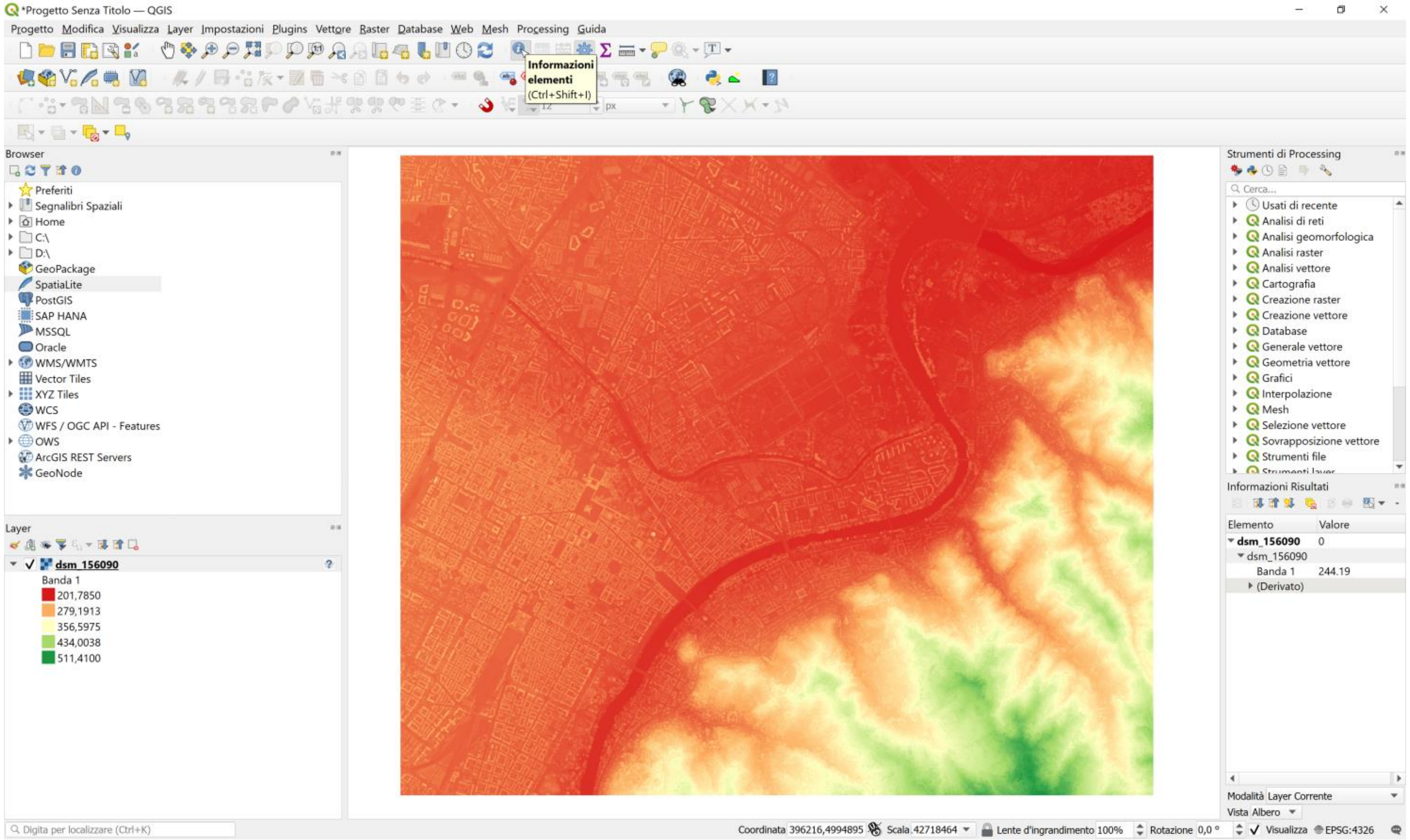
Ex 1


*Untitled Project — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Layer Properties — DTM_ritagl
Multiband color
Paletted/Unique values
Singleband gray
Singleband pseudocolor
Hillshade
Contours
Band Rende
Render type
Band
Band 1 (Gray)
Min
288,6799927
Max
534,2000122
Min / Max Value Settings
Interpolation
Linear
Color ramp
Label unit suffix
Label precision
4
Value
Color
Label
Mode
Continuous
Classes
5
Classify
Legend Settings...
Clip out of range values
Color Rendering
Blending mode
Normal
Reset
Brightness
0
Contrast
0
Gamma
1,00
Saturation
0
Grayscale
Off
Hue
Colorize
Strength
100%
Resampling
Zoomed: in
Nearest Neighbour
out
Nearest Neighbour
Oversampling
2,00
Early resampling
Style
OK
Cancel
Apply
Help
Information
Source
Symbology
Transparency
Histogram
Rendering
Temporal
Pyramids
Metadata
Legend
QGIS Server
Browser
Favorites
Spatial Bookmarks
Home
C:\
D:\
GeoPackage
SpatiaLite
PostGIS
SAP HANA
MSSQL
Oracle
WMS/WMTS
cartografia
ortofoto
Vector Tiles
XYZ Tiles
OpenStreetMap
WCS
Layers
DTM_ritaglio
Band 1 (Gray)
534,200012
288,679993

Ex 1


*Progetto Senza Titolo — QGIS
Progetto Modifica Visualizza Layer Impostazioni Plugins Vettore Raster Database Web Mesh Processing Guida
Browser
Strumenti di Processing
Layer
dsm_156090
Banda 1
511,410004
201,785004
201,7850
279,1913
356,5975
434,0038
511,4100

Ex 1



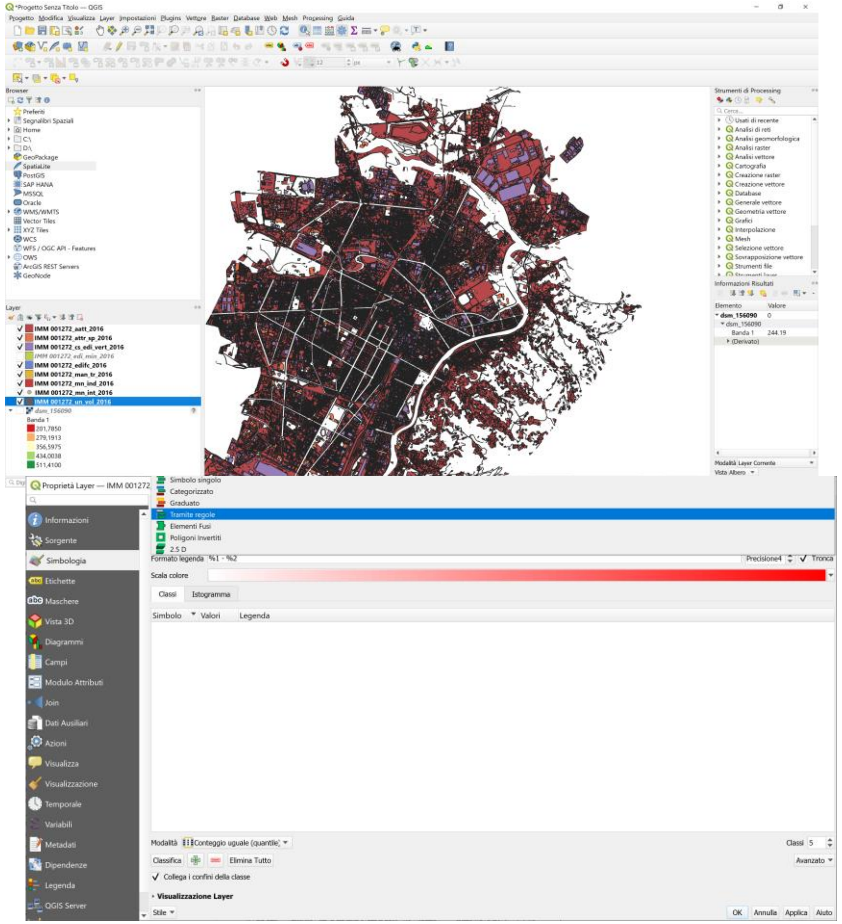
SELECTING TOOL «INFORMATION» it is possible to see the DIGITAL NUMBER
RASTER data doesn't have Attribute table

Ex 2

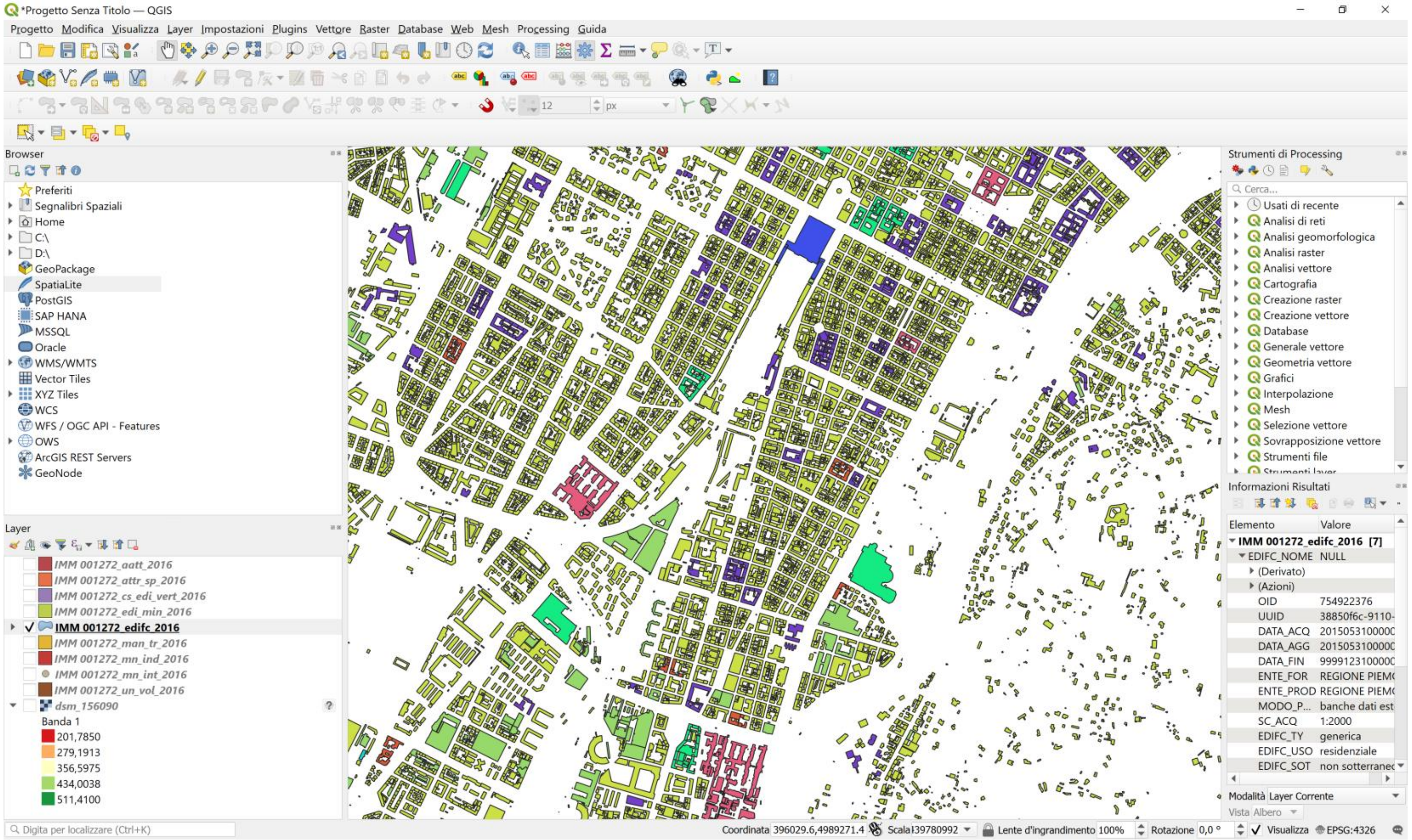
1) Extract the file in «BDTRE_DATABASE_GEOTOPOGRAFICO_2018-LIMI_COMUNI_10_GAIMSDWL-001028-EPSG32632-SHP.zip
2) Select «LAYER» → ADD LAYER → ADD VECTOR LAYER
3) Select the folder«IMM»
4) Importing only the layer «001028_edifc_2018»
5) Select the layer and right button, select properties

6) Select «symbology» and use «CATEGORIZED»
7) Select the field «EDIFC_USO» (use of building)

8) Click CLASSIFY and OK
9) Select the color ramp
10) Select ok



Ex 2

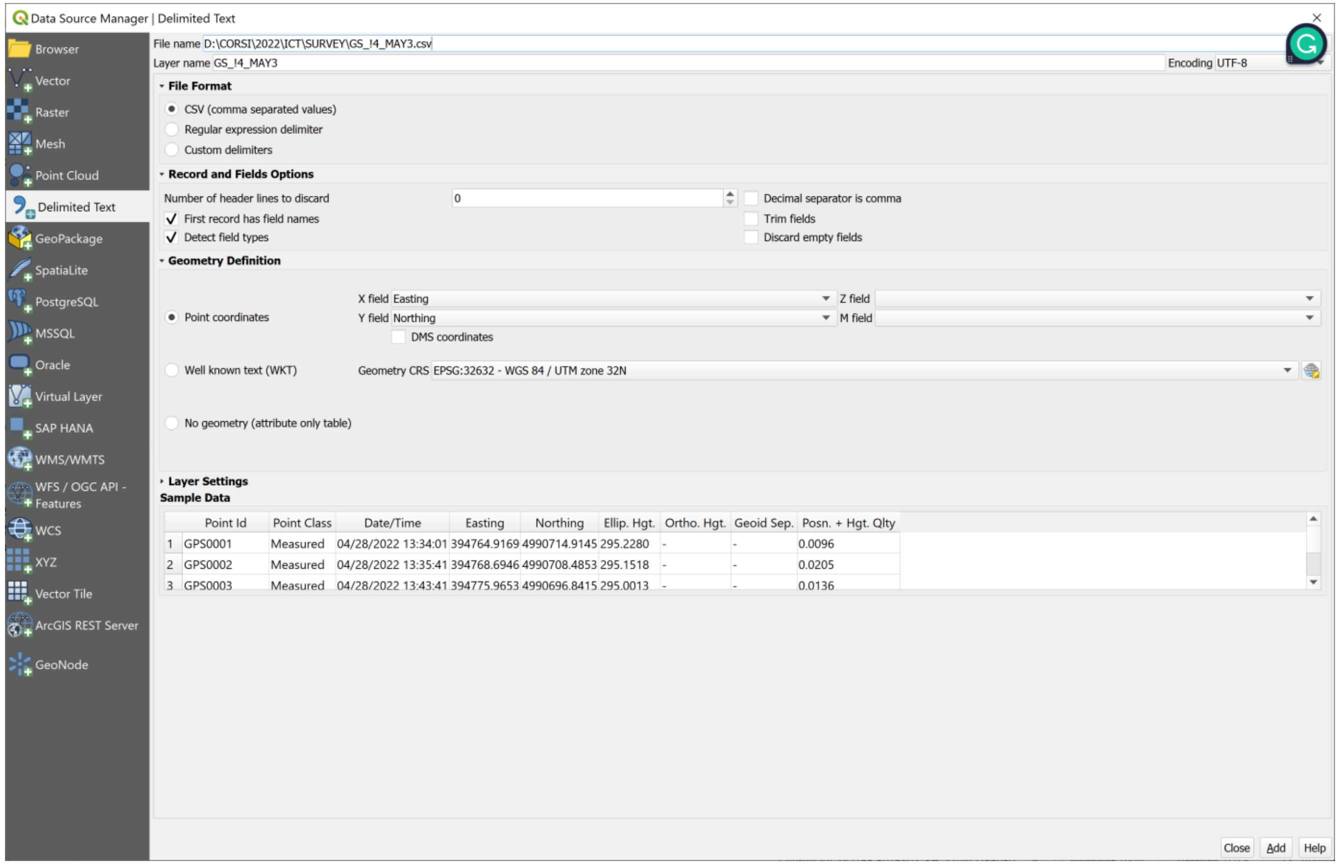

*Progetto Senza Titolo — QGIS
Progetto Modifica Visualizza Layer Impostazioni Plugins Vettore Raster Database Web Mesh Processing Guida
Browser
Preferiti
Segnalibri Spaziali
Home
C:\
D:\
GeoPackage
SpatiaLite
PostGIS
SAP HANA
MSSQL
Oracle
WMS/WMTS
Vector Tiles
XYZ Tiles
WCS
WFS / OGC API - Features
OWS
ArcGIS REST Servers
GeoNode
Layer
IMM 001272_aatt_2016
IMM 001272_attr_sp_2016
IMM 001272_cs_edi_vert_2016
IMM 001272_edi_min_2016
IMM 001272_edifc_2016
IMM 001272_man_tr_2016
IMM 001272_mn_ind_2016
IMM 001272_mn_int_2016
IMM 001272_un_vol_2016
dsm_156090
Banda 1
201,7850
279,1913
356,5975
434,0038
511,4100
Strumenti di Processing
Cerca...
Usati di recente
Analisi di reti
Analisi geomorfologica
Analisi raster
Analisi vettore
Cartografia
Creazione raster
Creazione vettore
Database
Generale vettore
Geometria vettore
Grafici
Interpolazione
Mesh
Selezione vettore
Sovrapposizione vettore
Strumenti file
Informazioni Risultati
Elemento Valore
IMM 001272_edifc_2016 [7]
EDIFC_NOME NULL
(Derivato)
(Azioni)
OID 754922376
DATA_ACQ
DATA_AGG
DATA_FIN
ENTE_FOR
ENTE_PROD
MODO_P...
SC_ACQ 1:2000
EDIFC_TY generica
EDIFC_USO residenziale
EDIFC_SOT
Modalità Layer Corrente
Vista Albero
Digita per localizzare (Ctrl+K)
Coordinata 396029.6,4989271.4
Lente d'ingrandimento 100%
Rotazione 0,0 °
Visualizza
EPSG:4326

Ex 3

IMPORT A TABLE

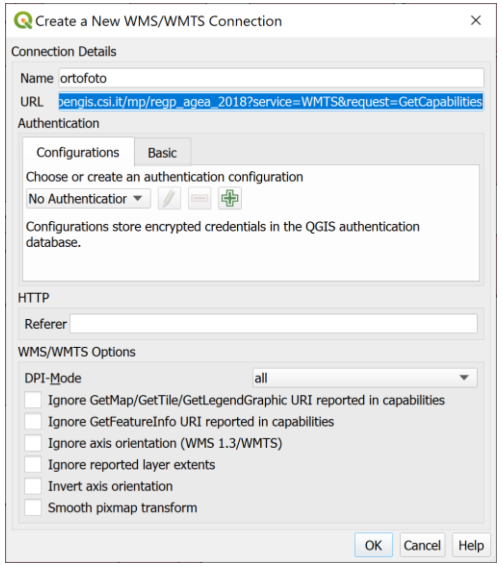
1) SELECT «LAYER» → ADD LAYER → ADD DELIMITED TEXT LAYER

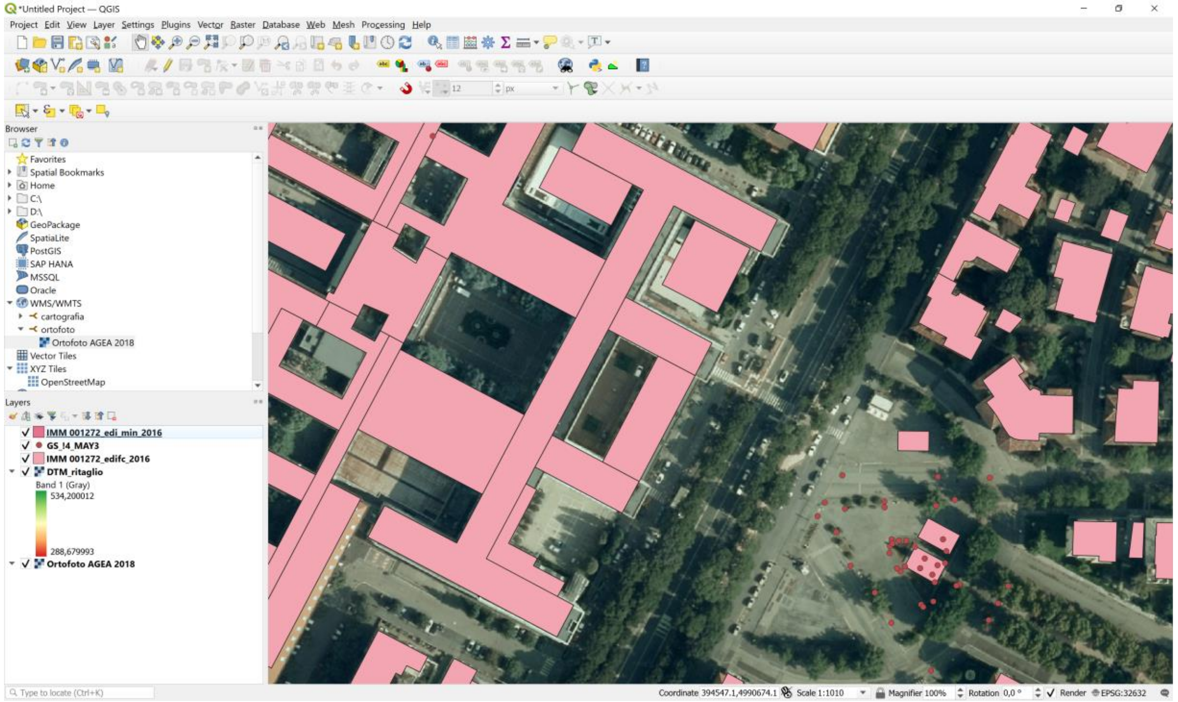
2) USE the CSV file in GNSS



IMPORT WMS

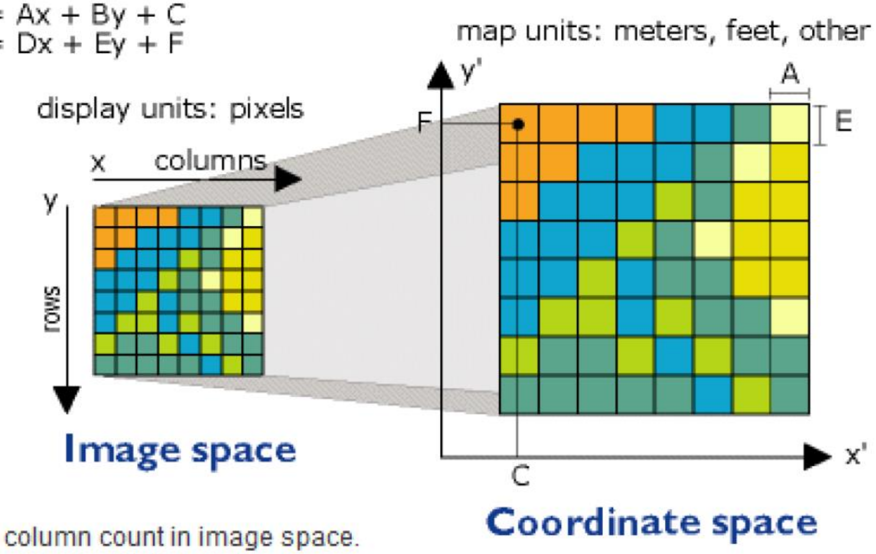
1) SELECT «LAYER» → ADD LAYER → ADD WMS





http://opengis.csi.it/mp/regp_agea_2018?service=WMTS&request=GetCapabilities

$$x' = Ax + By + C$$
$$y' = Dx + Ey + F$$



x is column count in image space.
y is row count in image space.
x' is horizontal value in coodinate space.
y' is vertical value in coordinate space.

A is width of cell in map units.
B is a rotation term.
C is the x' value of the center of the upper-left cell.
D is a rotation term.
E is negative height of cell in map units.
F is the y' value of the center of the upper-left cell.

Ex 5

GOAL:
GEOREFERENCING A RASTER

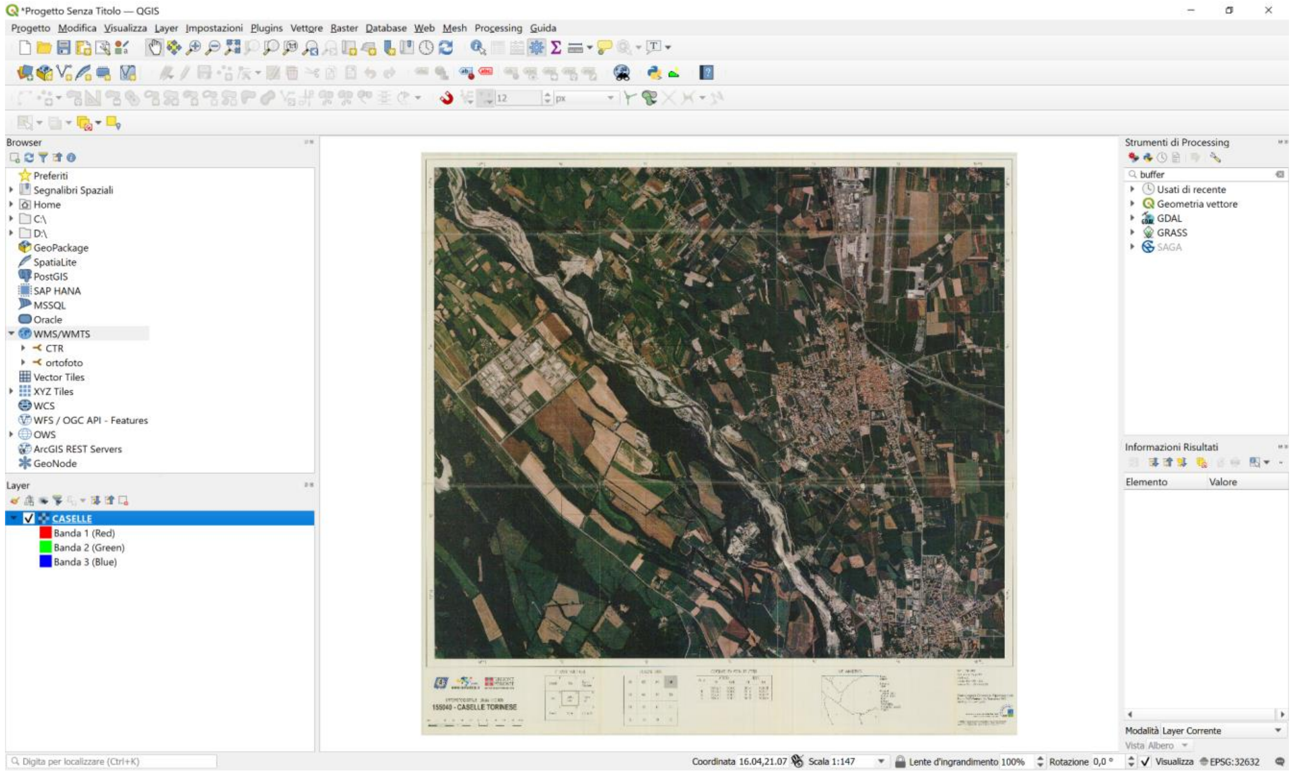
DATA:
1) CASELLE.TIF
2) VECTOR DATA about (BDTRE) (zip folder BDTRE….)

Ex 5

1) IMPORT the RASTER CASELLE.TIF

2) Image will be in «Pixel» coordinates

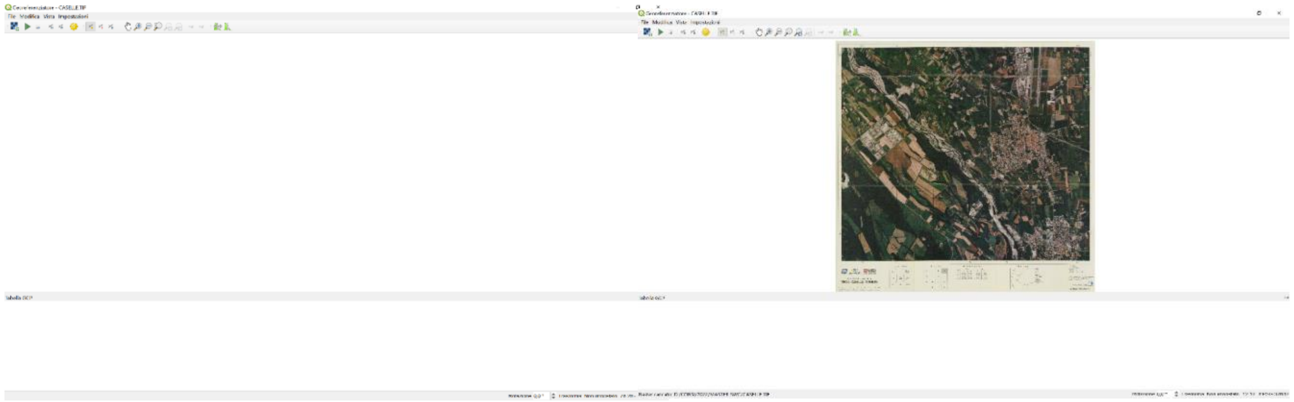
3) Select in Layer → GEOREFERENCER



Ex 5

1) IMPORT RASTER «CASELLE.TIF»

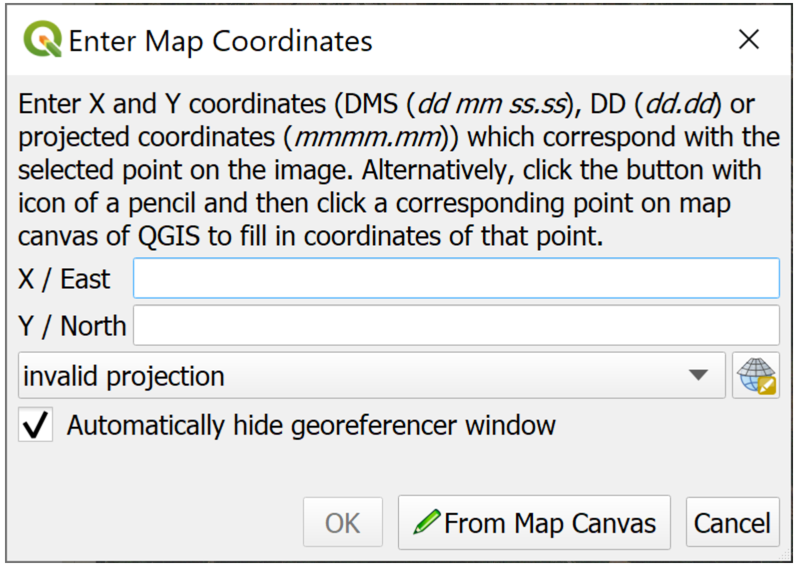
2) Start the collimation of the point



3) Select «add points…». Collimate the intersection of the grid

4) Insert the coordinates(East and North) that are on the external grid

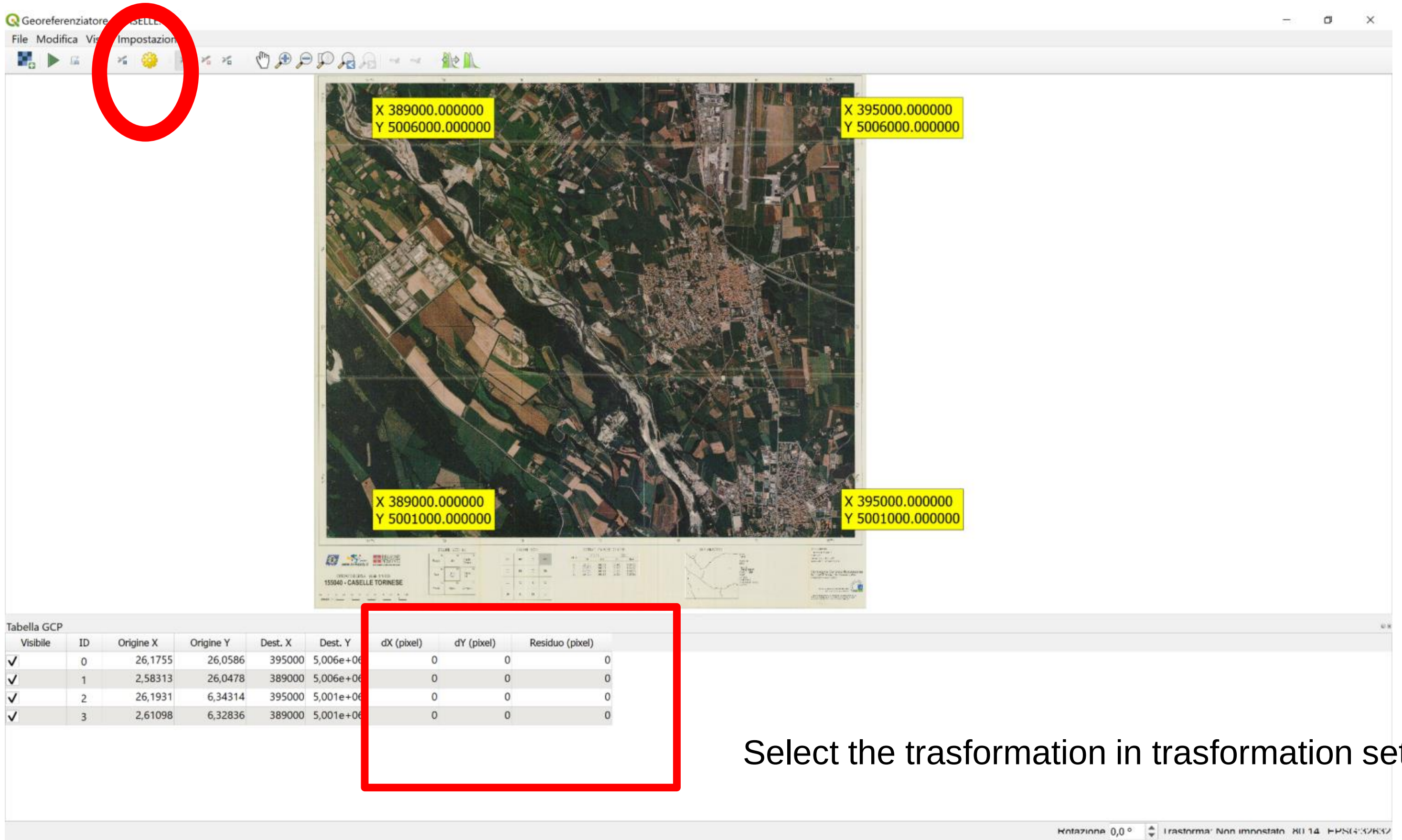
5) Collimate at least 4 points.



Ex 5

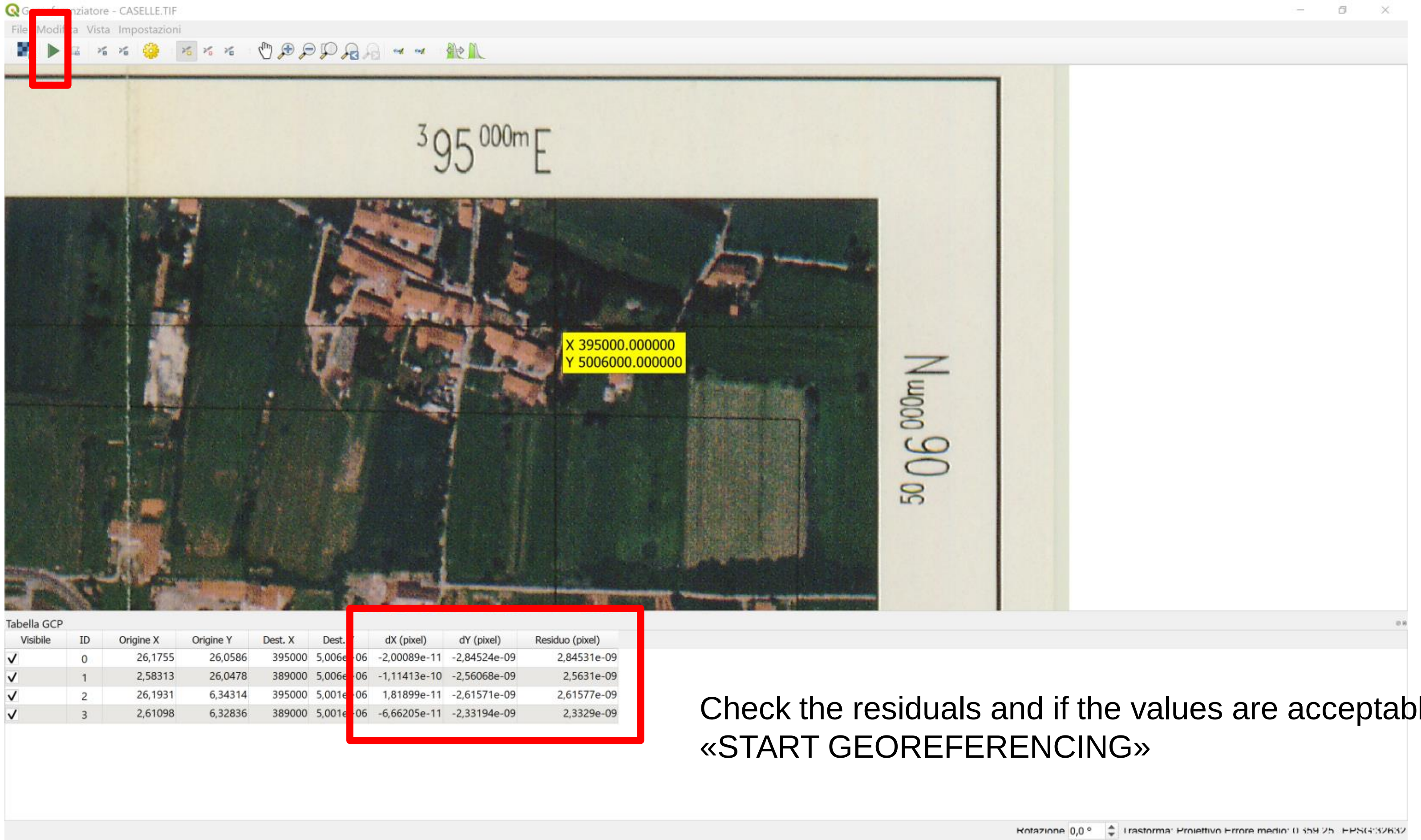

Georeferenziatore - CASELLE.TIF
File Modifica Vista Impostazioni
395 000m E
X 395000.000000
Y 5006000.000000
5006 000m N

Ex 5



Select the trasformation in trasformation setting

Ex 5

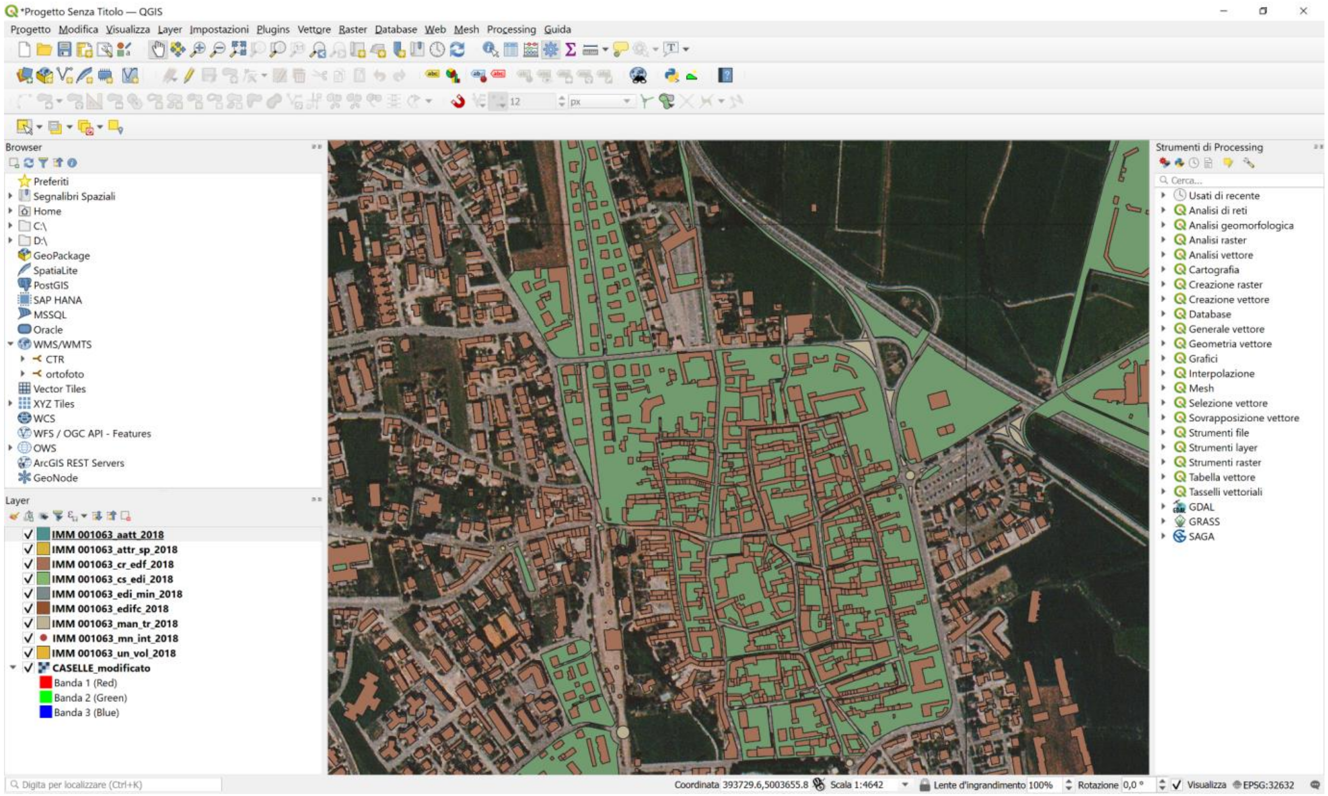


Check the residuals and if the values are acceptable, «START GEOREFERENCING»

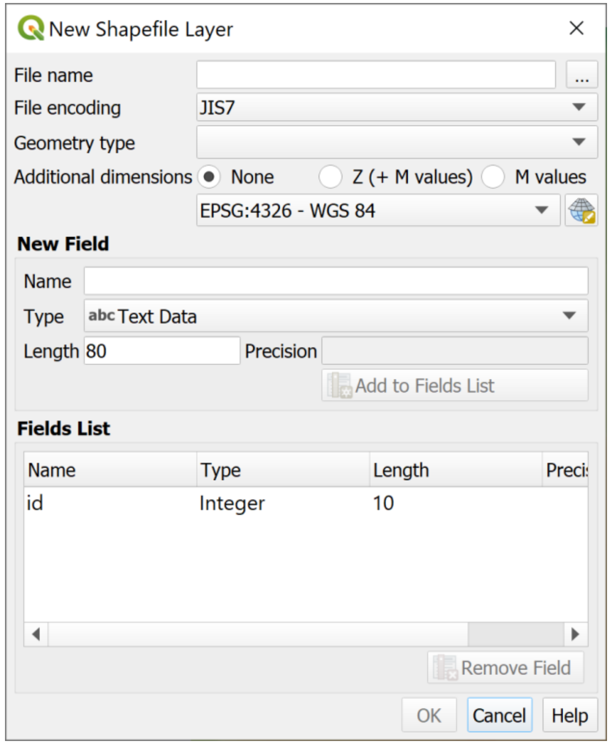
Ex 5

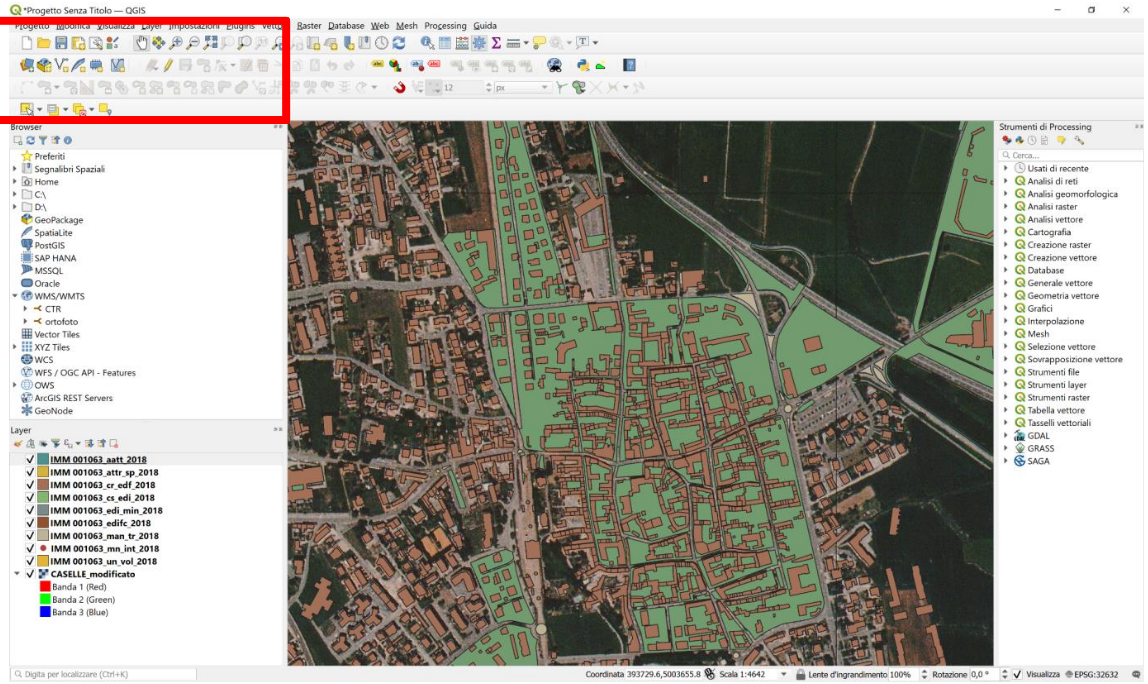
1) IMPORT THE BDTRE (just 1 folder)

2) Check the overlapping between map (raster) and vector data



CREATE NEW FEATURES (Entities)



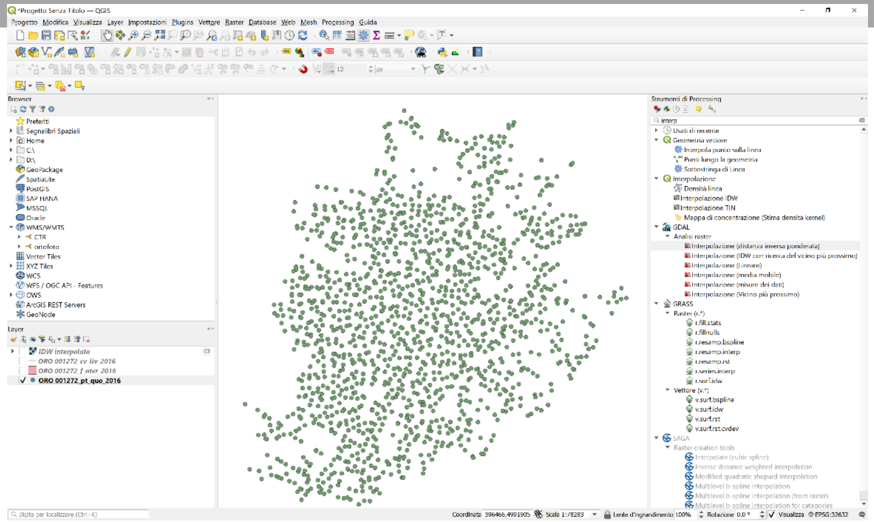


Ex 7

GOAL:
CREATE A DTM, using a point cloud selection

DATA:
1) FOLDER «ORO»

Ex 7

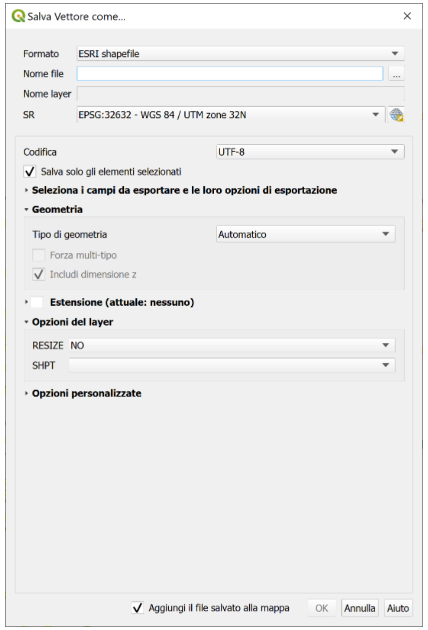


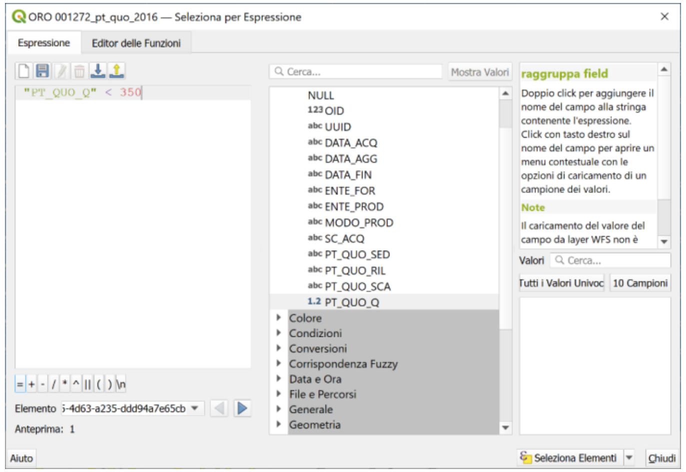
1) IMPORT THE FOLDER«ORO»

2) Select only the layer «pt_quo_2016»,

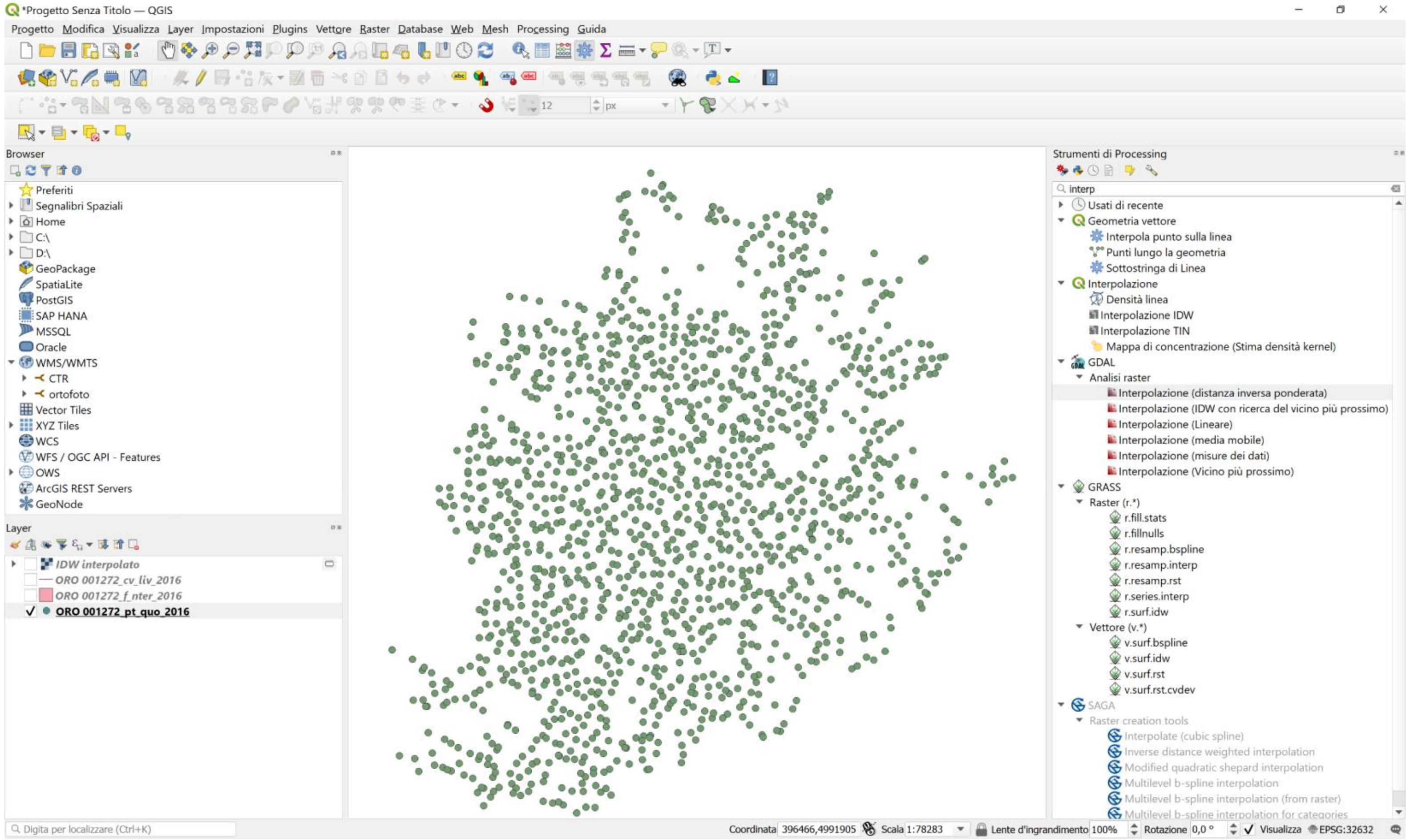
3) Select only the points with elevation < 350 m using **selection by attribute**

4) EXPORT THE SELECTION





Ex 7



*Progetto Senza Titolo — QGIS
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Browser
Preferiti
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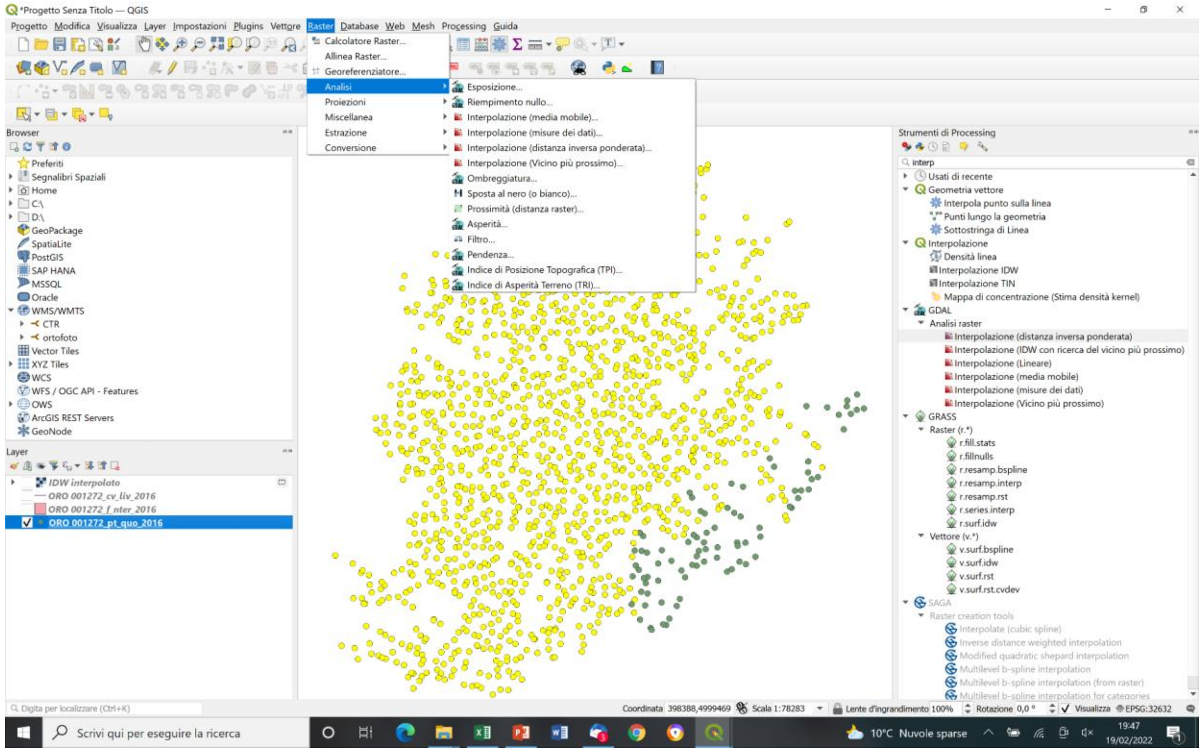
ortofoto
Vector Tiles
XYZ Tiles
WCS
WFS / OGC API - Features
OWS
ArcGIS REST Servers
GeoNode
Layer
IDW interpolato
ORO 001272_cv_liv_2016
ORO 001272_f_nter_2016
ORO 001272_pt_quo_2016
Strumenti di Processing
interp
Usati di recente
Geometria vettore
Interpola punto sulla linea
Punti lungo la geometria
Sottostringa di Linea
Interpolazione
Densità linea
Interpolazione IDW
Interpolazione TIN
Mappa di concentrazione (Stima densità kernel)
GDAL
Analisi raster
Interpolazione (distanza inversa ponderata)
Interpolazione (IDW con ricerca del vicino più prossimo)
Interpolazione (Lineare)
Interpolazione (media mobile)
Interpolazione (misure dei dati)
Interpolazione (Vicino più prossimo)
GRASS
Raster (r.*)
r.fill.stats
r.fillnulls
r.resamp.bspline
r.resamp.interp
r.resamp.rst
r.series.interp
r.surf.idw
Vettore (v.*)
v.surf.bspline
v.surf.idw
v.surf.rst
v.surf.rst.cvdev
SAGA
Raster creation tools
Interpolate (cubic spline)
Inverse distance weighted interpolation
Modified quadratic shepard interpolation
Multilevel b-spline interpolation
Multilevel b-spline interpolation (from raster)
Multilevel b-spline interpolation for categories
Digita per localizzare (Ctrl+K)
Coordinata 396466,4991905 Scala 1:78283 Lente d'ingrandimento 100% Rotazione 0,0 ° Visualizza EPSG:32632

Ex 8

1) Using the IDW and NN function, create 2 DTM
Starting by point cloud



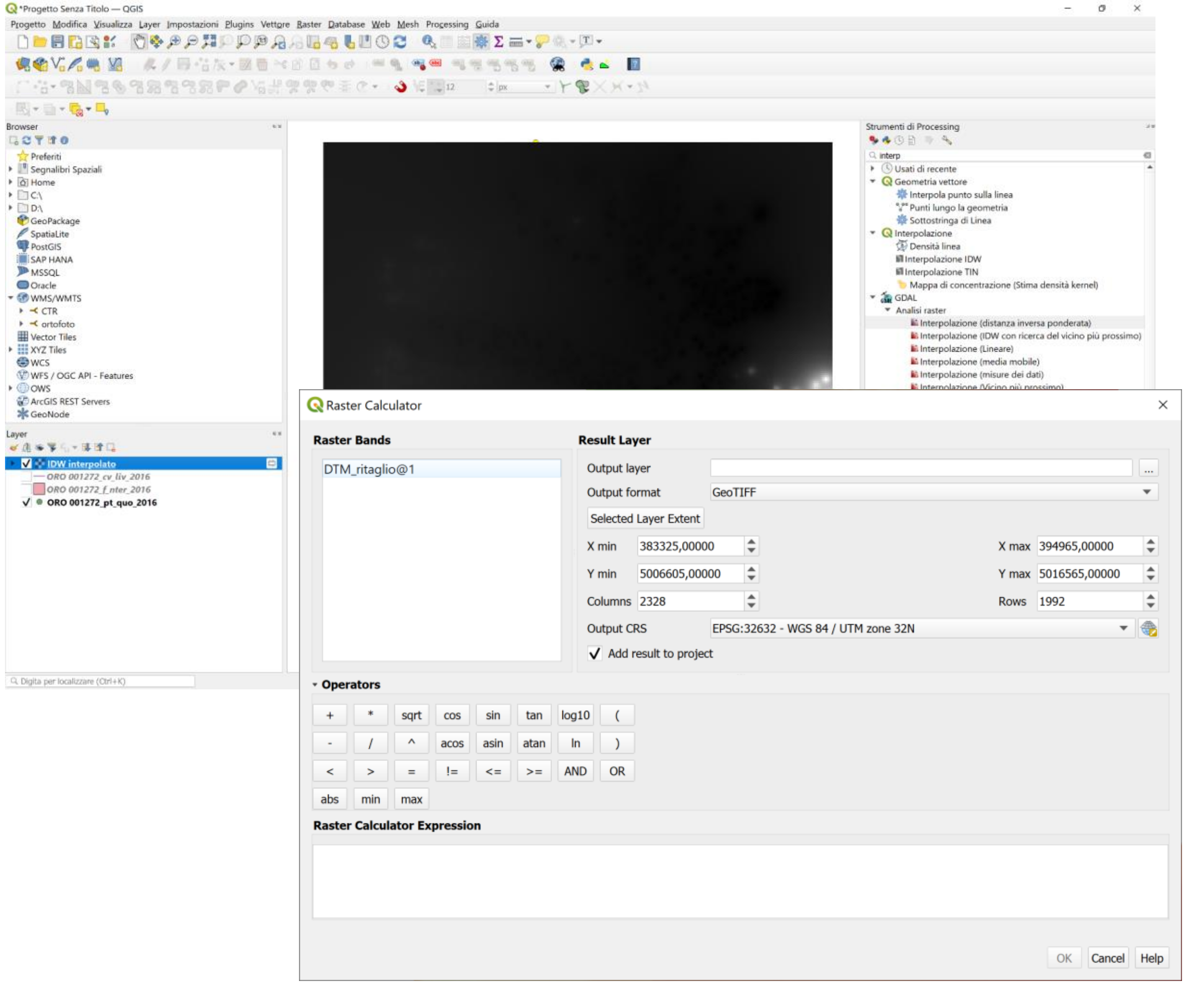
Ex 8

1) USING RASTER CALCULATOR, making the differences

RASTER → raster calculator

Select
RASTER IDW - RASTER NN

Result will be a new raster



Ex 9

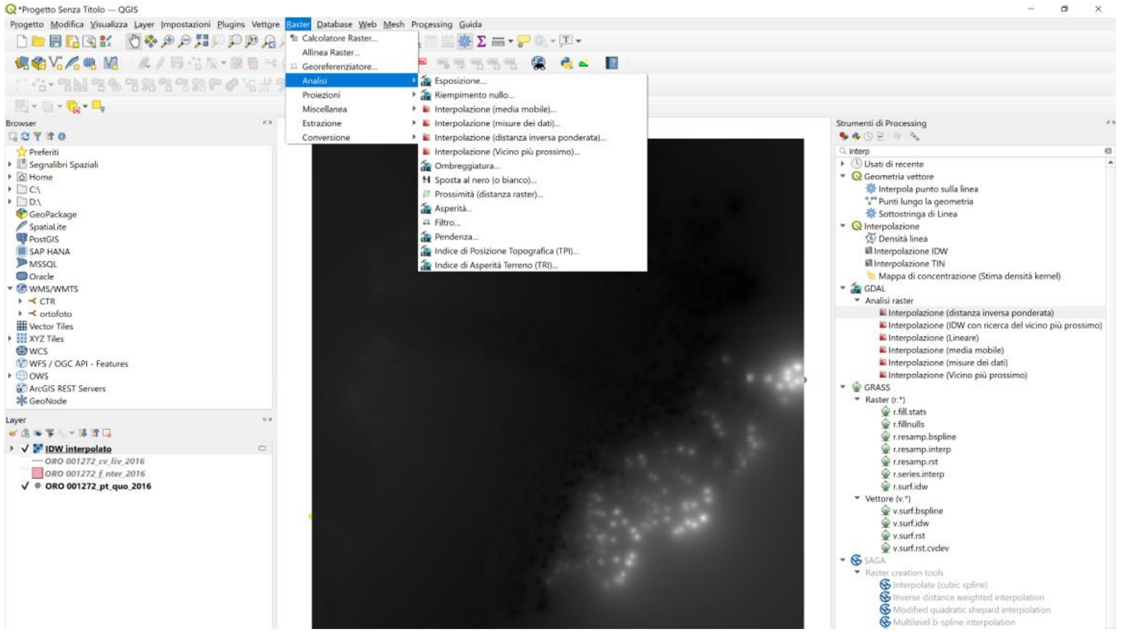
GOAL:
Using a DTM, to extract contour lines, slope map, aspect map.

DATI:
1) DTM 155120.asc

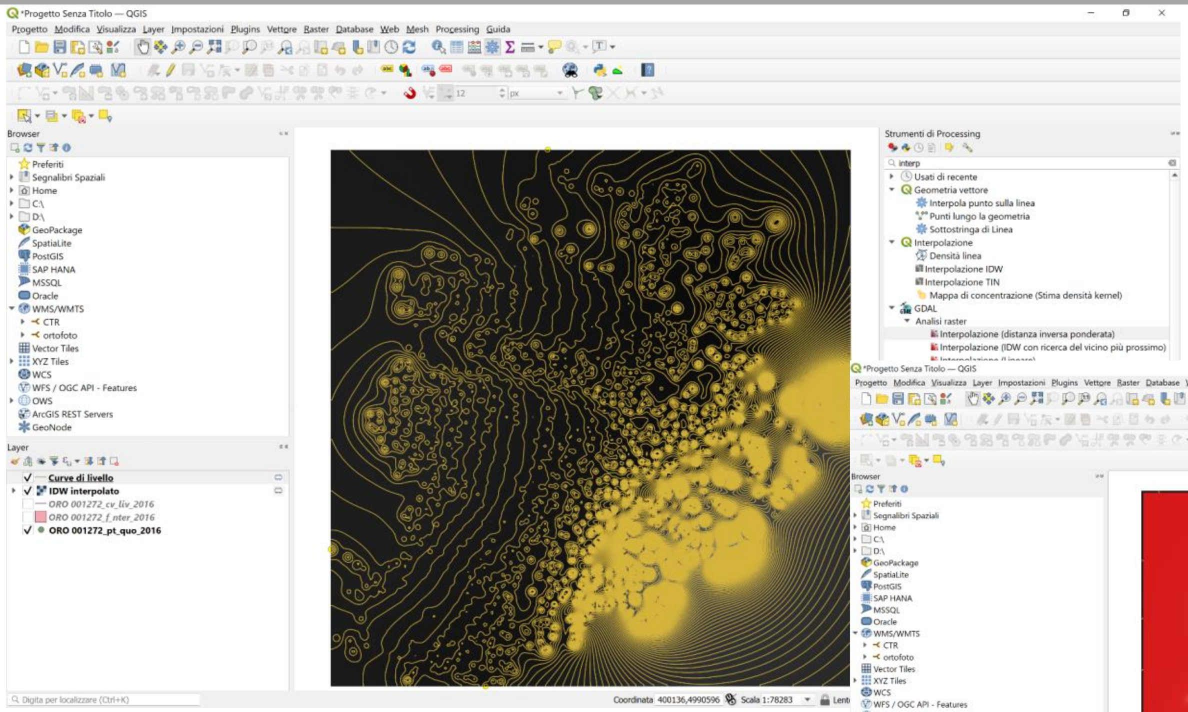
Ex 9

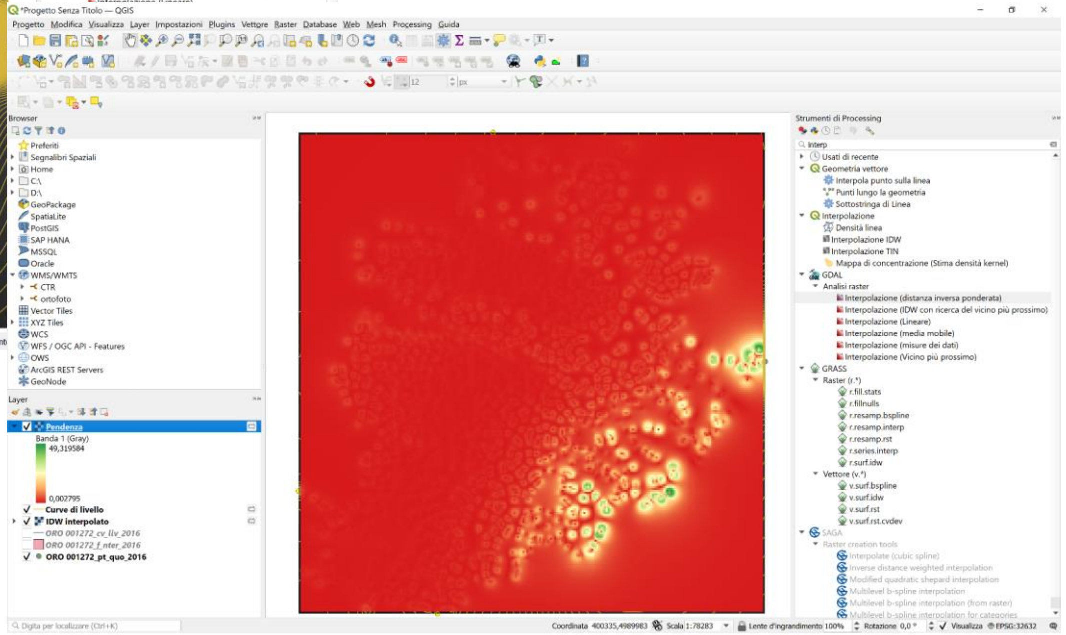
1) IN RASTER → Analysis

2) IN RASTER → Extraction
For contour lines
slope map, aspect map.



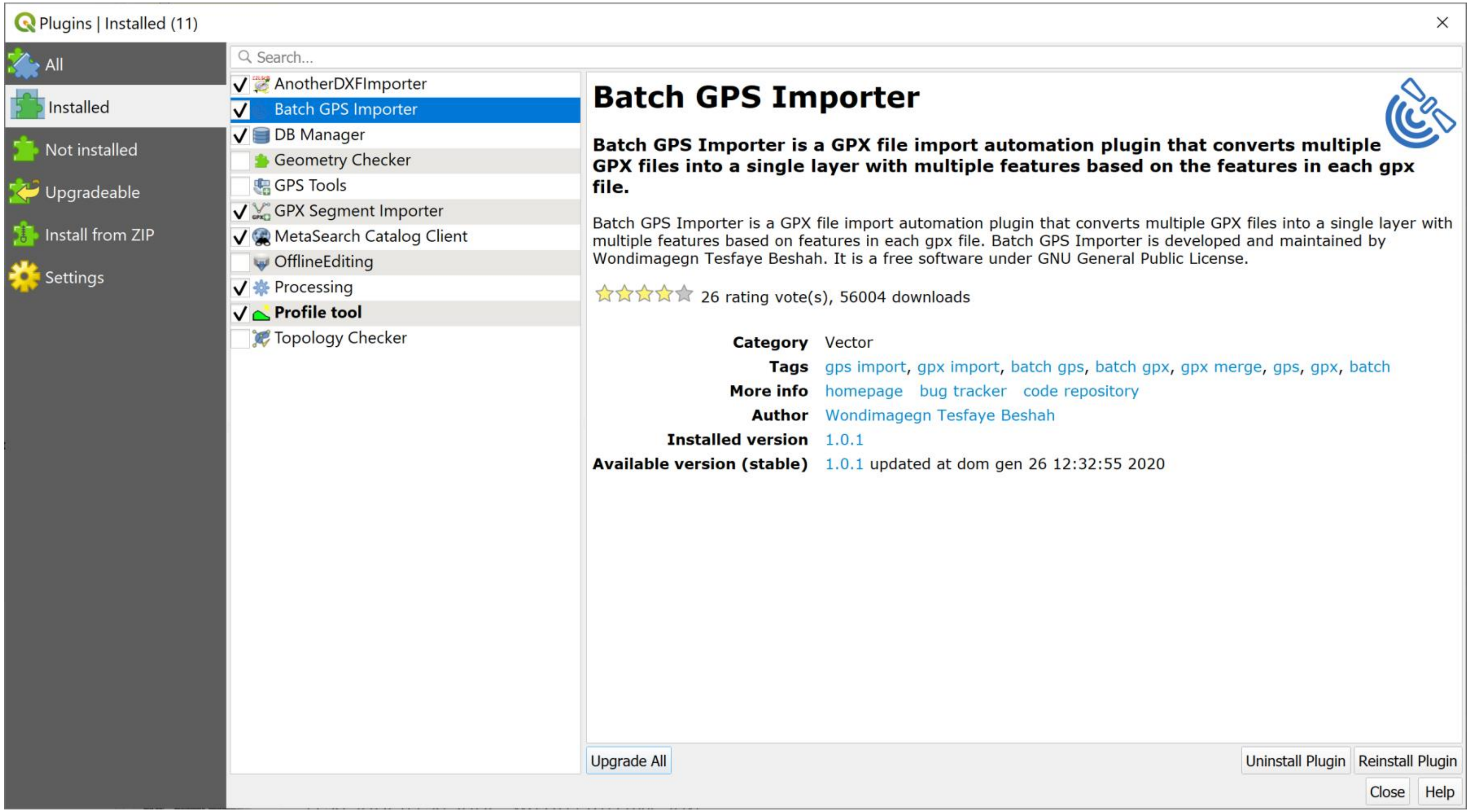
Ex 9





Change the symbology using previous EX

Ex 10 – import GPX



Ex 11 – create a layout